Jeet Bahadur Thapa
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Dayton, Ohio | 937-414-7125 (Open to Relocation | Flexible: Remote, Hybrid, or On-site)

EDUCATION:

- **Master of Science in Mechanical Engineering** — Jan 2024 – Present
  Wright State University, Dayton, Ohio — Cumulative GPA: 3.857 / 4.0

SOFTWARE AND TECHNICAL SKILLS:

- *CAD/Simulation*: SolidWorks, Simufact, ANSYS, MATLAB, Python
- *Testing Equipment*: Formlabs Fuse 1 (SLS printer), FDM printer, Preform, Prusca Slicer, Tensile test (UTS), MTS (ASTM D695 compression test), Keyence VR6300 (roughness), Mitutoyo Vision System (dimension), ImageJ, SEM, XRD, DSC, TGA, TMA
- *Vibration Analysis:* Spectral Dynamics Bobcat analyzer, RMS/time-domain/frequency-domain analysis

CERTIFICATIONS, AWARDS & RESEARCH PRESENTATIONS:

- CSWA, CSWP, CSWA-AM, CSWPA-DT, Oasis Rapid Certification on Semiconductor & Integrated manufacturing
- Conferences for Poster Presenter for 3D printing (SLS) and Polymer characterization:
  - MS&T 2025, DESS 2025, PINO 2026

PROFESSIONAL EXPERIENCE:

Graduate Research Assistant:

Additive Manufacturing & Characterization: *Wright State University, Ohio* *May 2025 – Present*

- Performed ASTM D695 compression tests on SLS-printed PA12/PA12-GF at rates of 0.01–5 s^{-1} (MTS 244.51) and conducted ISO 179-1 Charpy impact tests (Instron Ceast 9050) to measure modulus, yield, stress–strain response, and toughness across different build orientations. Additionally, performed metrology and surface characterization using Mitutoyo Vision System & Keyence VR-6000 to assess Sa/Sq roughness & topography.
- Completed XRD analysis on PA12 powder versus printed parts for crystalline phase comparison and SEM fractography on PA12-GF after cryogenic fracture to evaluate fracture morphology & glass bead microstructure.
- Conducted DSC to determine Tm, Tc, and ΔHm for melting and crystallization behaviors, and TGA to generate mass loss and degradation onset curves, benchmarking thermal stability of PA12 versus PA12-GF.

Graduate Teaching Assistant (GTA): `

Mechanical Vibrations Lab: *Wright State University, Ohio* *Aug 2025 – Dec 2025*

- Conducted SDOF vibration experiments (cantilever beam & forced vibration) to evaluate stiffness, damping, natural frequency, and mode shapes; guided FRF and transmissibility analysis using accelerometers, impulse hammers, and rotating unbalance excitation — linking experimental results to vibration theory

Mechanical and Material Testing Lab: *Wright State University, Ohio* *Jan 2025 – April 2025*

- Tensile Test (Instron), Elastic Beam Bending, Heat Treatment Effect, Impact Test (Charpy), Fatigue Test

Technical Advisor & Sr. Senior Engineer: *B & K Enterprises, Nepal* *Dec 2018 – Nov 2023*

- Implemented standardized vehicle service SOPs across multiple vehicle classes, improving shop throughput and saving 225+ labor hours/year; led component failure analysis/RCA (hydraulics, clutch, brake systems) and preventive actions reducing repeat failures by ~15%.
- Delivered technical operations support through warranty/inventory reporting (Microsoft Office, Microsoft Dynamics Navision), team coordination for cost/quality targets, and emissions compliance inspections using exhaust gas analyzers (gasoline) and opacity meters (diesel)

Service Advisor: *Sipradi Trading Pvt. Ltd., Nepal* *Jul 2016 – Dec 2018*

- Improved service turnover by ~20% by streamlining diagnostics using TATA Diagnostics Software (DTCs, live data, actuator tests, wiring checks) and implementing structured job cards/repair planning.
- Deployed Hunter Hawkeye Elite 3D alignment to improve camber/caster/toe accuracy with post-service verification, and trained technicians on updated OEM procedures after TATA Motors HQ programs.

PROJECTS:

- Retrofitting of a 2-stroke Gasoline Engine to a Compressed Air Engine — *Nov 2014 – Nov 2015*
- 3D Analysis of the AGARD 445.6 Wing by FEA for Vibration & Flutter Prediction — *Oct 2024- Nov 2024*
- Simulation of Metal Forming to Analyze the Effects of Geometry and Friction Using Simufact — *2025*